scientific reports

OPEN

A hybrid machine learning model for intrusion detection in wireless sensor networks leveraging data balancing and dimensionality reduction

Md. Alamin Talukder[1]✉, Majdi Khalid[2]✉ & Nasrin Sultana[3]✉

Intrusion detection systems are essential for securing wireless sensor networks (WSNs) and Internet of Things (IoT) environments against various threats. This study presents a novel hybrid machine learning (ML) model that integrates KMeans-SMOTE (KMS) for data balancing and principal component analysis (PCA) for dimensionality reduction, evaluated using the WSN-DS and TON-IoT datasets. The model employs classifiers such as Decision Tree Classifier, Random Forest Classifier (RFC), and gradient boosting techniques like XGBoost (XGBC) to enhance detection accuracy and efficiency. The proposed hybrid (KMS + PCA + RFC) approach achieves remarkable performance, with an accuracy of 99.94% and an f1-score of 99.94% on the WSN-DS dataset. For the TON-IoT dataset, it achieves 99.97% accuracy and an f1-score of 99.97%, outperforming traditional SMOTE TomekLink and Generative Adversarial Network-based data balancing techniques. This hybrid approach addresses class imbalance and high-dimensionality challenges, providing scalable and robust intrusion detection. Complexity analysis reveals that the proposed model reduces training and prediction times, making it suitable for real-time applications.

Keywords Intrusion detection systems, Wireless sensor networks, Internet of Things, Hybrid machine learning, Model, Dimensionality reduction, Data balancing techniques

Internet of Things (IoT) networks and wireless sensor networks (WSNs) have become pivotal in modern technology, playing a crucial role in numerous fields, including industrial automation, smart cities, agriculture and healthcare[1]. WSNs consist of spatially distributed autonomous sensors that monitor and record environmental conditions, while IoT networks enable interconnected devices to communicate and exchange data seamlessly[2]. Despite their potential, these networks are vulnerable to numerous security threats, including data tampering, denial-of-service (DoS) attacks, routing attacks, and eavesdropping, which can severely compromise their performance and reliability[3,4]. The dynamic and resource-constrained nature of WSNs and IoT networks presents unique security challenges[5]. Traditional methods for intrusion detection often fall short due to their inability to adapt to the heterogeneous and evolving attack patterns in these networks[6,7]. Over the years, several machine learning (ML)-based approaches have been proposed for intrusion detection systems (IDS) in WSNs and IoT networks[8,9]. These approaches leverage the capability of ML to learn complex patterns, enabling automated detection of anomalies and malicious activities with improved accuracy compared to rule-based systems[10,11].

ML has emerged as a powerful tool in IDS due to its ability to automatically identify complex patterns and anomalies in network data[12,13]. ML models can effectively detect known and unknown attacks by learning from historical data, offering dynamic and adaptive solutions[14–17]. Techniques like Decision Trees (DT), Random Forests (RF), and Gradient Boosting classifiers provide high accuracy, making them ideal for real-time threat detection in WSN and IoT environments[18].

[1]Department of Computer Science and Engineering, International University of Business Agriculture and Technology, Dhaka, Bangladesh. [2]Department of Computer Science and Artificial Intelligence, College of Computing, Umm Al-Qura University, 21955 Mecca, Saudi Arabia. [3]Department of ICT, Future Technology, RMIT University, Melbourne, Australia. ✉email: alamin.cse@iubat.edu; mknfiai@uqu.edu.sa; nasrin.sultana@rmit.edu.au

Existing works in the wireless sensor and IoT network domain have demonstrated promising results using various ML models such as DT, usfAD, one class classifier (OCC) and Gradient Boosting models[19–22] to build the IDS model. However, these methods often suffer from limitations, including imbalanced datasets and high dimensionality of features, which degrade model performance. For instance, imbalanced datasets lead to biased predictions favoring majority classes, while high-dimensional data increases computational costs and risks overfitting.

To address these limitations, we propose a novel hybrid ML model that integrates KMeans-SMOTE (KMS) for data balancing with Principal Component Analysis (PCA) for dimensionality reduction, and Random Forest Classifier (RFC) as the core attack detection model. The proposed model mitigates the issues of data imbalance and feature redundancy while maintaining high accuracy and efficiency. By employing KMS, the proposed approach effectively balances minority and majority classes, ensuring equitable representation in the dataset. PCA reduces the dimensionality of data, capturing essential features while eliminating noise, thereby enhancing computational efficiency. Finally, the use of RFC ensures robust and interpretable decision-making.

Our proposed methodology surpasses the limitations of existing techniques by offering a more balanced, efficient, and scalable solution for intrusion detection in WSN and IoT networks. Through extensive experimentation on WSN-DS and TON-IoT datasets, our model demonstrates superior performance in terms of accuracy, precision, recall, and f1-score, thereby providing a robust framework for securing these critical networks against evolving threats.

Objective

The primary objective of this research is to develop a robust hybrid ML model for intrusion detection in WSNs. By integrating advanced data balancing techniques such as STL, GANs, and KMS, and reducing the curse of dimensionality using PCA, the model aims to address the challenges of class imbalance and high-dimensional data. This approach enhances detection accuracy, minimizes false positives, and ensures computational efficiency, making it suitable for real-time applications in WSNs security.

Contributions

This study offers the following important contributions:

- **Novel Hybrid Machine Learning Approach:** This study introduces a hybrid approach combining data balancing and dimensionality reduction techniques, specifically tailored for intrusion detection in Wireless Sensor Networks. It ensures improved accuracy and robustness by addressing class imbalance and high-dimensional data challenges.
- **Advanced Data Balancing Techniques:** The integration of data balancing techniques such as STL, GANs, and KMS, can effectively resolve overfitting and underfitting issues in imbalanced WSN datasets. These methods enhance minority class representation and improve the model's ability to detect intrusions accurately.
- **Dimension Reduction with PCA:** PCA reduces the curse of dimensionality by projecting data onto significant features using the Kaiser criterion, thereby improving computational efficiency and enhancing classification accuracy. This technique also minimizes noise and irrelevant data, making the model scalable and reliable for real-time intrusion detection.This paper is systematized as follows: "Section Related works" presents an overview of related works, highlighting the recent works in intrusion detection systems. "Section Methodology" details the proposed methodology, including preprocessing and model development. "Section Result analysis" analyzes the experimental results analysis, followed by a discussion comparing with the existing works in "Section Discussion". Finally, "Section Conclusion" concludes the study, addressing its limitations and proposing directions for future research.

Related works

IDS using ML have gained significant attention for securing WSNs. Various ML models have been applied to IDS for detecting abnormal activities. These models effectively classify network traffic and identify intrusions, though challenges such as class imbalance and high-dimensional data often hinder their performance. Recent advancements incorporate hybrid techniques, ensemble learning, and deep learning (DL) to improve detection accuracy and robustness.

Related works on WSN-DS

Talukder et al.[23] introduced an IDS method that combined ML with the STL algorithm. This approach balanced the dataset by generating synthetic minority instances and removing Tomek links, leading to improved detection accuracy in WSNs. To ensure consistent and scalable input features, feature scaling through standardization was applied. The STL resampling technique effectively addressed imbalanced datasets, reducing the risks of overfitting and underfitting. An extensive evaluation using the WSN-DS dataset revealed the RFC model, achieving 99.92% accuracy, precision, recall, and f1-score, in multiclass scenarios, demonstrating its effectiveness in identifying and preventing intrusions.

Karthikeyan et al.[20] made significant advancements in IoT and WSN security by combining ML with the Firefly Algorithm (FA). This work has two key contributions: first, the FA-ML technique enhances intrusion detection accuracy in WSN-IoT systems, and second, it introduces a novel security optimization approach by integrating ML with the Firefly Algorithm. The FA-ML method, utilizing an SVM model and optimized by the Grey Wolf Optimizer (GWO), achieved an impressive performance with 99.34% accuracy, 98.36% recall and 96.67% f1-score. These results demonstrate the FA-ML method's effectiveness as a robust security solution for WSN-IoT systems.

Abinayaa et al.[24] introduced an innovative approach to intrusion detection in WSN by combining the CatBoost classifier (Cb-C) with the Lyrebird Optimization Algorithm (LOA). Cb-C effectively handles imbalanced datasets commonly found in ID settings. Inspired by the Lyrebird's ability to mimic environmental sounds, the LOA is a metaheuristic optimization algorithm. The proposed method was evaluated using the WSN-DS dataset. Among the models tested, LOA-Cb-C achieved the highest accuracy of 99.66% and 99% of precision, recall and f1-score and the lowest error rate of 0.34%, significantly improving WSN-IoT security by enhancing detection accuracy and reducing false alarms.

Salmi et al.[25] developed an effective IDS to address intelligent and frequent attacks in WSNs, particularly Denial of Service (DoS) attacks. Traditional IDS methods were deemed less effective against these complex threats. The study reviewed works focused on detecting DoS attacks in WSNs and implemented several DL-based IDS. These systems were trained on the WSN-DS dataset to detect four types of DoS attacks: Blackhole, Grayhole, Flooding, and Scheduling. The CNN+RNN model achieved significant performance with 85.19% precision, 85.29% recall, 82.34% f1-score, and 96.86% accuracy.

Dontu et al.[26] developed a multi-criteria framework to strengthen WSN cybersecurity, utilizing DL algorithms to prioritize threats according to their severity and context. The study introduced the Enhanced Adaptive Butterfly Optimization Algorithm (EABOA) for feature selection, addressing challenges such as slow convergence, vulnerability to local optima, and lack of population diversity. Additionally, a new adaptive fragrance model was proposed to enhance algorithm performance, incorporating Lévy flight for better exploration and escape from local optima. The experimental results showed that the model on WSN-DS achieved an accuracy of 94.76%, precision of 86%, recall of 80%, and an F-score of 80.06%.

Vaishnavi et al.[27] conducted a study on detecting Denial of Service (DoS) attacks in WSNs using ML techniques. The research addresses the challenges of DoS attacks in WSN environments, using the WSN-DS dataset for analysis. It applies data preprocessing methods like PCA and Min-Max Scaling and employs Convolutional Neural Networks (CNN) and XGBoost algorithms in various combinations. Among these models, XGBoost with Min-Max Scaling achieved the highest accuracy of 99.70%, outperforming the other approaches.

Related works on TON-IoT-DS

Cao et al.[28] developed an innovative approach to simulate complex IoT scenarios by integrating multiple datasets. This method employed sine and cosine cyclic encoding for temporal attributes and addressed data imbalance using (STL) SMOTE for oversampling and an improved Tomek link removal for under-sampling. ML algorithms were utilized to classify data, distinguishing between normal and attack instances. Multiclass classification identified specific attack types, enriching the attack database, while normal data was stored securely for future use. Evaluation demonstrated exceptional performance, achieving 98.87%, 99.36%, 99.18%, and 99.18% for accurac, precision, recall, and F1 score respectively.

Uddin et al.[21] developed a semi-supervised learning-based IDS that does not require labeled attack data. The approach consisted of two methods: (1) training a supervised model using randomly generated synthetic attack samples, and (2) creating a One-Class Classification (OCC) model using only benign network traffic. Both methods were tested on benchmark IDS datasets. The proposed usfAD model achieved 99.43% accuracy, 98.39% precision, and 99.37% f1-score on the TON-IoT Network dataset. Results demonstrated that the OCC model, utilizing the usfAD anomaly detection technique, outperformed conventional supervised and other OCC-based methods, particularly in detecting new, unseen attacks.

Hajla et al.[29] proposed an improved intrusion detection system (IDS) framework to strengthen cybersecurity in IoT networks. Departing from traditional IDS methods that relied on predefined attack patterns, this framework focused on multiclass attack categorization using the TON-IoT dataset. It employed STL to balance the dataset and ensemble techniques such as voting and stacking, which integrated decision trees, resilient backpropagation multilayer perceptrons (RProp MLPs), and logistic regression to enhance accuracy and reduce errors. The ensemble voting model outperformed individual classifiers, achieving 96.32% accuracy, 93.11% precision, 84.55% recall, and 88.63% f1-score, demonstrating its effectiveness in detecting IoT cyber threats.

Li et al.[30] analyzed feature extraction and selection techniques for IoT network intrusion detection using the TON-IoT dataset. Feature extraction demonstrated superior detection accuracy, greater feature retention, and stability with varying feature counts, while feature selection excelled in reducing model training and inference time. Decision trees (DT), using PCA with 33 features, achieved 77.62% accuracy, 67.34% precision, 48.67% recall, and a 45.53% f1-score. The research highlighted the strengths and limitations of each approach, offering insights into selecting effective methods for IoT intrusion detection and addressing prior gaps in the field.

Torre et al.[19] developed a federated learning (FL) IDS utilizing a 1D CNN to enhance IoT network security. To address privacy concerns in FL, the system incorporated Differential Privacy, Diffie-Hellman Key Exchange, and Homomorphic Encryption. The approach was evaluated on the TON-IoT dataset, achieving exceptional results with an accuracy of 99.63%, precision of 96.48%, recall of 93.18%, and an f1-score of 94.45%. These outcomes demonstrated the model's effectiveness in accurately detecting intrusions within IoT networks while maintaining data privacy.

Uddin et al.[22] introduced a two-tier, one-class classification-based IDS. The first tier differentiates between normal behavior and attacks, while the second tier identifies known and unknown attacks using a multi-classification mechanism and clustering algorithm. This allows the system to recognize and adapt to new, unseen threats by clustering unknown attacks and retraining. By employing one-class classifiers (OCC), the model eliminates the need for attack samples, tackling issues like data imbalance and zero-day attacks. The proposed framework demonstrated strong performance, achieving an accuracy and f1-score of 98.94% on the TON-IoT network dataset, showing its practical deployment potential.

Methodology

This study introduces a comprehensive methodology designed to enhance the intrusion detection performance of ML models on the WSN and TON-IoT Network datasets. The methodology integrates advanced data preprocessing, dimensionality reduction, and ML approaches, ensuring optimal accuracy and computational efficiency. The steps involved are as follows:

- **Data Collection:** The study utilizes two benchmark datasets, WSN-DS and TON-IoT Network, to evaluate the proposed hybrid machine learning model for intrusion detection. These datasets contain diverse traffic patterns and attack scenarios, providing a comprehensive foundation for training and testing the model's performance across different network conditions.
- **Data Preprocessing:** Data preprocessing is a crucial step in preparing the dataset for ML models. StandardScaler was used to standardize the numerical feature values by removing the mean and scaling to unit variance, which helps in achieving faster convergence during training. Label Encoding was applied to transform categorical variables into numerical representations, enabling ML algorithms to process them effectively. These preprocessing techniques ensure uniformity, reduce bias, and enhance the overall performance of the models.
- **Data Balancing:** The dataset was imbalanced, with certain classes being overrepresented. To address this issue, SMOTE TomekLInk (STL), Generative Adversarial Network (GAN) and KMeans-SMOTE (KMS) were utilized to balance the dataset by generating synthetic samples for minority classes. KMS leverages k-means clustering before applying SMOTE, ensuring better synthetic data generation by capturing intrinsic patterns in the data distribution.
- **Dimensionality Reduction:** Principal component analysis (PCA) was employed as the dimensionality reduction technique. PCA reduces the feature space by projecting the data into a lower-dimensional subspace, retaining the most significant variance. This step minimizes redundancy and computational cost while preserving the critical information needed for intrusion detection.
- **Classification Models:** Random Forest Classifier (RFC) was chosen as the primary classification model due to its robustness, ability to handle high-dimensional data, and resistance to overfitting. The RFC model operates by constructing multiple decision trees during training and combining their outputs to improve prediction accuracy. For comparison, other models, including Decision Tree Classifier (DTC), Gradient Boosting models (CBC, XGBC, LGBC, HGBC), were also evaluated.
- **Performance Metrics:** The intrusion detection performance of the models was evaluated using standard metrics such as Accuracy, Precision, Recall, f1-score, Kappa, MCC, ROC-AUC. MAE, MSE, RMSE, Confusion Matrix and Classification Report. Additionally, computational complexity was assessed by recording training and prediction times for each model.
- **Experimental Validation:** The proposed methodology, integrating KMS, PCA, and RFC, was compared against existing works utilizing different data balancing processing and dimension reduction techniques with K-fold (K=5) cross validation. The results demonstrated that the proposed approach significantly outperforms others in terms of performance metrics and computational efficiency, validating its effectiveness for IoT network anomaly detection tasks.This end-to-end methodology ensures balanced data, reduced dimensionality, and high intrusion detection accuracy, making it well-suited for real-world applications in IoT network security.

Figure 1 illustrates the architecture of the proposed hybrid ML model for IDS in WSNs. The model integrates advanced preprocessing, dimensionality reduction, and robust intrusion detection techniques to enhance detection accuracy and computational efficiency. It highlights the end-to-end workflow, from raw data processing to intrusion detection and performance evaluation of attacks in WSN.

Dataset description

The WSN-DS (Wireless Sensor Network Dataset) is a specialized benchmark for evaluating IDS in WSN environments[31]. It simulates realistic WSN traffic, including both normal and malicious activities, such as blackhole attacks, grayhole attacks, and flooding. The dataset is designed to reflect the constraints and characteristics of WSNs, such as limited bandwidth, energy consumption, and dynamic topology. WSN-DS provides a balanced representation of normal and attack behaviors, facilitating the development and testing of ML-based IDS models. Its features include packet-level data and communication patterns, enabling detailed analysis for intrusion detection. The dataset is critical for designing IDS tailored to the unique challenges of WSNs, ensuring security while maintaining network efficiency and resource optimization.

The TON-IoT Network dataset is a prominent benchmark designed to evaluate Intrusion Detection Systems (IDS) in modern IoT networks[32]. It is derived from realistic IoT network scenarios, incorporating telemetry data from IoT devices, network traffic, and logs from operating systems. The dataset includes a variety of attack types, such as Distributed Denial of Service (DDoS), data exfiltration, backdoor, and password attacks, alongside normal traffic, making it highly diverse and comprehensive. The dataset is notable for its high dimensionality and heterogeneity, reflecting the dynamic and complex nature of IoT networks. It offers an extensive range of features, enabling the evaluation of ML models for intrusion detection. TON-IoT Network dataset is a critical resource for researchers aiming to enhance the scalability, robustness, and accuracy of IDS models in addressing the security challenges of large-scale IoT ecosystems.

Data preprocessing

Data preprocessing is a critical step in the ML pipeline, involving the preparation and transformation of raw data into a suitable format for analysis and modeling. This process encompasses a series of techniques aimed at

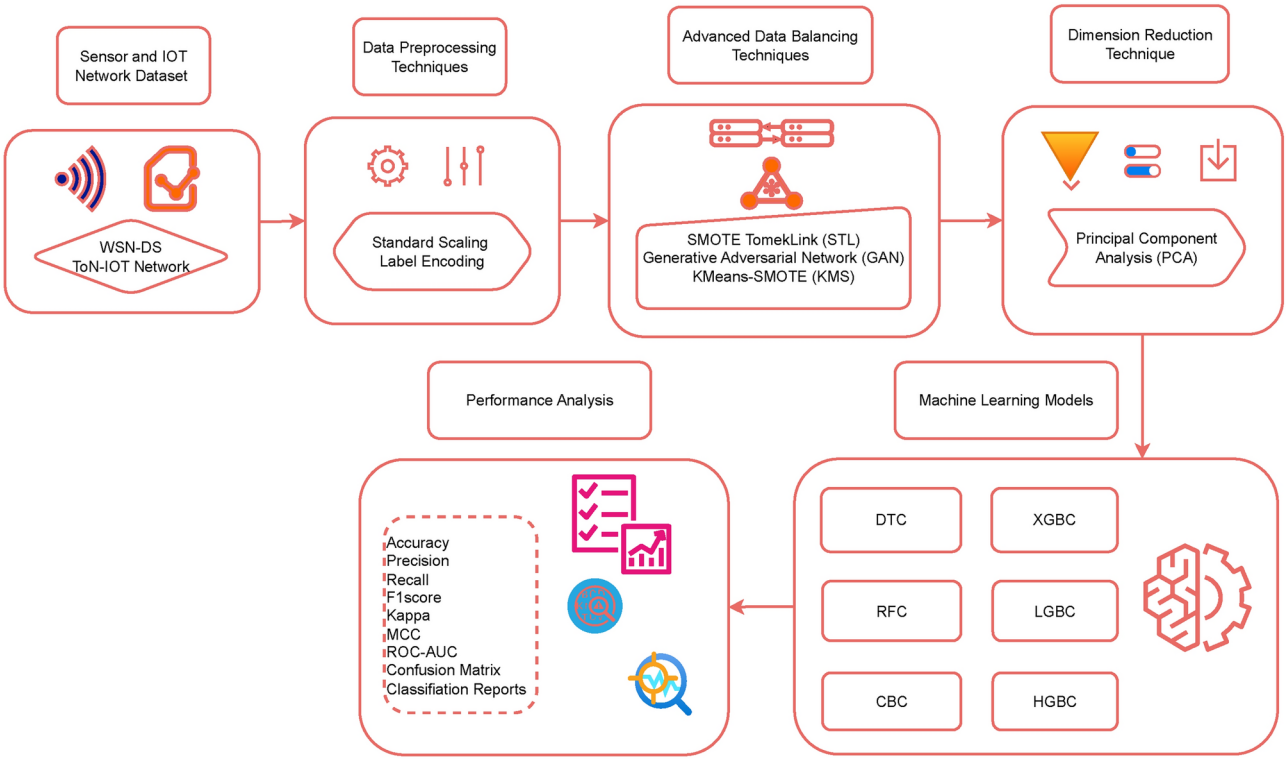

Fig. 1. The proposed hybrid ML model for IDS in WSNs.

cleaning, normalizing, transforming, and structuring data to enhance its quality and usability. The preprocessing techniques utilized in this study include StandardScaler and Label Encoding. StandardScaler was applied to standardize the feature values by removing the mean and scaling to unit variance, ensuring uniformity across features and improving model convergence. Label Encoding was employed to convert categorical variables into numerical format, facilitating seamless integration with ML algorithms while preserving the inherent order of categorical data. It is essential to mitigate inconsistencies, redundancies, and noise in the dataset, which can negatively impact model performance. Without proper preprocessing, ML models may produce biased or inaccurate results, especially when handling imbalanced or high-dimensional data.

StandardScaler

The StandardScaler is a preprocessing technique used to normalize features by removing the mean and scaling to unit variance. It ensures that the features have a mean of 0 and a standard deviation of 1, making the data well-suited for ML algorithms sensitive to feature magnitudes. The standardization of a feature x is computed as:

$$z = \frac{x - \mu}{\sigma} \tag{1}$$

where x: Original feature value; μ: Mean of the feature; σ: Standard deviation of the feature.

StandardScaler improves the model by ensuring that features are on a comparable scale, thereby allowing ML algorithms to converge faster and avoid biases due to varying feature magnitudes. Additionally, it enhances numerical stability, preventing issues arising from extreme values. The StandardScaler is particularly beneficial for distance-based models, such as clustering algorithms, where feature scaling directly impacts the calculation of distances.

Label encoding

Label Encoding is a preprocessing technique used to convert categorical labels into numerical values. Each unique category is assigned a specific integer value, enabling ML algorithms to process non-numerical labels effectively.

Let y represent the categorical labels. The encoding maps each unique category c to a corresponding integer value i:

$$f(c) = i, \quad \text{where } i \in \{0, 1, \ldots, n-1\} \tag{2}$$

Here, n is the total number of unique categories, and $f(c)$ is the mapping function.

Category	Label
Normal	0
Grayhole	1
Blackhole	2
TDMA	3
Flooding	4

Table 1. Category-to-label mapping.

Category	Label
Normal	0
Scanning	1
DoS	2
Injection	3
DDoS	4
Password	5
XSS	6
Ransomware	7
Backdoor	8
MITM	9

Table 2. Category-to-label mapping.

Label Encoding simplifies the representation of categorical labels, enabling compatibility with ML models that require numerical input. It is computationally efficient and facilitates the direct integration of categorical data into IDS model-building workflows. Tables 1 and 2 display the process of encoding categorical labels to numerical labels for different classes in the WSN-DS and TON-IoT Network Datasets.

Data balancing (DB) techniques

Data Balancing (DB) techniques aim to address the class imbalance problem in datasets by ensuring an even distribution of instances across all classes. This is crucial for improving the performance of ML models, particularly in IDS, where the occurrence of attacks (minority classes) is often much rarer than normal behavior (majority class). Data balancing techniques work by generating new samples for the minority classes or by modifying the dataset to reduce bias toward majority classes. These methods either synthesize new data points or combine existing ones to create balanced datasets.

The class imbalance ratio can be calculated as:

$$\mathrm{IR} = \frac{\text{Size of Majority Class}}{\text{Size of Minority Class}} \tag{3}$$

where a high imbalance ratio (IR) indicates severe class imbalance. DB techniques aim to minimize this ratio.

Class imbalance leads to biased models that perform poorly on minority classes, which are often the critical classes for intrusion detection. Balancing the dataset ensures fair learning, reduces overfitting, and improves the recall and precision of IDS models. Balancing techniques significantly enhance the accuracy and robustness of IDS models by addressing the imbalance issue. They help prevent overfitting to the majority class, enabling the model to detect rare but critical intrusion patterns more effectively.

STL—SMOTETomekLink

SMOTETomekLink (STL) is a hybrid approach combining the Synthetic Minority Oversampling Technique (SMOTE) and Tomek Link undersampling to handle both overfitting and class overlap issues in imbalanced datasets. SMOTE generates synthetic samples for the minority class by interpolating between existing instances. Tomek Link identifies and removes overlapping samples from majority and minority classes, enhancing data separation.

SMOTE generates new samples using:

$$\mathrm{x}_{new} = \mathrm{x}_i + \delta \cdot (\mathrm{x}_j - \mathrm{x}_i) \tag{4}$$

where x_i and x_j are minority class samples, and δ is a random number between 0 and 1.

STL addresses both the imbalance and noise in datasets by combining oversampling and undersampling, reducing the risk of model overfitting and misclassification. STL enhances IDS models' recall and precision by generating realistic synthetic samples while eliminating noisy instances, improving overall detection rates.

GAN—generative adversarial network

Generative Adversarial Networks (GANs) are DL-based methods that generate synthetic samples for the minority class using a generator and discriminator model[33]. The generator creates synthetic samples, while the discriminator evaluates their authenticity against real samples. The generator learns to produce realistic samples that the discriminator cannot distinguish from real data.

The optimization goal for GAN is:

$$\min_{G} \max_{D} \mathbb{E}_{x \sim P_{data}}[\log D(x)] + \mathbb{E}_{z \sim P_z}[\log(1 - D(G(z)))] \tag{5}$$

where G is the generator, D is the discriminator, x represents real data, and z is a noise vector.

GANs are effective in generating complex and realistic data distributions, making them suitable for highly imbalanced datasets. GANs create high-quality synthetic samples for minority classes, improving the ability of IDS models to detect rare intrusions without overfitting.

KMS—KMeansSMOTE

KMeansSMOTE is a clustering-based oversampling technique that applies K-Means clustering to partition data before generating synthetic samples using SMOTE. The dataset is first clustered using K-Means, and SMOTE is applied within each cluster to generate synthetic samples for the minority class. This localized oversampling preserves the original data distribution and reduces noise.

The clustering is performed as:

$$\arg\min_{C} \sum_{i=1}^{k} \sum_{x \in C_i} ||x - \mu_i||^2 \tag{6}$$

where C represents clusters, μ_i is the centroid of cluster i, and k is the number of clusters.

KMS overcomes the limitations of global oversampling by focusing on local distributions within each cluster, generating more representative synthetic samples. KMS improves the balance and quality of IDS datasets, leading to more accurate and robust intrusion detection. By preserving data distribution, it reduces the risk of overfitting and enhances detection rates for minority classes. In this study, we propose using KMS for data balancing. Its ability to cluster data before generating synthetic samples ensures that the data distribution is preserved, and noise is minimized. This approach makes it particularly well-suited for improving the performance of IDS models.

Table 3 highlights the impact of data balancing techniques-STL, GAN, and KMS-on the WSN-DS dataset. It demonstrates how these methods address class imbalances by significantly increasing the number of samples in minority classes, ensuring a balanced dataset for effective model training. Notably, KMS achieves comparable balancing performance with slightly optimized total samples, enhancing data consistency.

Table 4 demonstrates the effects of data balancing techniques-STL, GAN, and KMS-on the TON-IoT network dataset. The table highlights how these methods substantially increase the number of samples in minority classes, ensuring a balanced distribution across various attack types. KMS shows an effective performance in data balancing, providing a slight improvement in the total number of samples compared to STL and GAN while maintaining a well-distributed dataset for accurate model training.

The integration of KMS addresses the challenges of class imbalance in WSNs intrusion datasets by clustering the majority class data using KMeans and generating synthetic samples within these clusters for the minority

STL			GAN			KMS		
Class	**Before**	**After**	**Class**	**Before**	**After**	**Class**	**Before**	**After**
Normal	340066	340063	Normal	340066	340066	Normal	340066	340068
Grayhole	14596	340038	Grayhole	14596	340066	Grayhole	14596	340067
Blackhole	10049	340020	Blackhole	10049	340066	Blackhole	10049	340066
TDMA	6638	339803	TDMA	6638	340066	TDMA	6638	340066
Flooding	3312	339784	Flooding	3312	340066	Flooding	3312	340066
Total	374661	1699708	Total	374661	1700330	Total	374661	1700333

Table 3. Before and after data balancing techniques on WSN-DS.

STL			GAN			KMS		
Class	**Before**	**After**	**Class**	**Before**	**After**	**Class**	**Before**	**After**
Normal	300000	300000	Normal	300000	300000	Normal	300000	300006
Scanning	20000	299999	Scanning	20000	300000	Scanning	20000	300005
Dos	20000	299997	Dos	20000	300000	Dos	20000	300004
Injection	20000	299995	Injection	20000	300000	Injection	20000	300003
Ddos	20000	299987	Ddos	20000	300000	Ddos	20000	300003
Password	20000	299980	Password	20000	300000	Password	20000	300003
Xss	20000	299979	Xss	20000	300000	Xss	20000	300003
Ransomware	20000	299975	Ransomware	20000	300000	Ransomware	20000	300002
Backdoor	20000	299966	Backdoor	20000	300000	Backdoor	20000	300001
Mitm	1043	299946	Mitm	1043	300000	Mitm	1043	300000
Total	461043	2999824	Total	461043	3000000	Total	461043	3000030

Table 4. Before and after data balancing techniques on TON-IoT network.

class. This approach ensures that the synthetic samples are representative of the distribution of the majority class while maintaining the integrity and diversity of the dataset. Compared to traditional balancing techniques like STL, KMS has a distinct advantage in preserving data integrity. While STL combines oversampling and under-sampling to address the imbalance and remove overlapping samples, it may inadvertently discard useful data, leading to the potential loss of information. In contrast, KMS avoids aggressive under-sampling by focusing on targeted oversampling within clusters, which helps retain the original dataset's structure and diversity. Furthermore, our experimental results demonstrate that KMS provides a balanced representation of classes, leading to improved detection accuracy and reduced false positives in intrusion detection tasks. This makes it a more suitable choice for datasets with complex distributions and varying levels of class imbalance.

Dimension reduction using PCA

Principal Component Analysis (PCA) is a statistical technique used for reducing the dimensionality of a dataset by transforming it into a new coordinate system, where the axes (principal components) represent directions of maximum variance in the data. PCA works by finding the eigenvectors and eigenvalues of the covariance matrix of the data. The eigenvectors define the principal components, and the corresponding eigenvalues measure the variance along these components. PCA transforms the data into a lower-dimensional space by selecting a subset of components that capture the most variance.

The covariance matrix is computed as:

$$\mathrm{C} = \frac{1}{n}\sum_{i=1}^{n}(\mathrm{x}_i - \mu)(\mathrm{x}_i - \mu)^T \tag{7}$$

where x_i is a data point, μ is the mean vector, and n is the number of samples.

Eigenvalues and eigenvectors are computed as:

$$\mathrm{C}\mathrm{v}_i = \lambda_i \mathrm{v}_i \tag{8}$$

where λ_i is an eigenvalue, and v_i is the corresponding eigenvector.

PCA reduces the curse of dimensionality in WSN datasets, enabling faster and more efficient ML model training. It eliminates redundant features and noise, improving the generalization ability of IDS models in wireless sensor networks. By focusing on key variations in the data, PCA also enhances model interpretability and reduces computational overhead.

Kaiser criterion

The Kaiser Criterion is a rule for selecting the number of principal components in PCA. It states that only components with eigenvalues greater than 1 should be retained, as these components explain more variance than an individual feature in the original dataset.

Process: 1. Compute the eigenvalues of the covariance matrix after applying PCA. 2. Retain only those components whose eigenvalues are greater than 1. 3. Transform the data using the selected components.

The number of components selected by the Kaiser Criterion is given by:

$$n_{\text{components}} = \sum_{i=1}^{k} \mathbb{1}(\lambda_i > 1) \tag{9}$$

where λ_i is the eigenvalue of the i-th component, k is the total number of components, and $\mathbb{1}(\cdot)$ is the indicator function.

The Kaiser Criterion ensures that only the most significant principal components are retained, reducing computational complexity without compromising the model's performance. For WSN datasets, this approach helps in efficiently capturing key variations, enhancing the accuracy of IDS models while maintaining scalability in resource-constrained environments.

The Kaiser Criterion is widely used because it provides a simple yet effective heuristic to select components that contribute meaningfully to the variance. For IDS datasets, the high-dimensional nature and the need to focus on features with significant explanatory power make eigenvalue ≥ 1 a reasonable threshold.

PCA efficiently reduces the high-dimensional feature space while retaining the most significant information. By eliminating redundant and less informative features using kaiser criterion, PCA minimizes noise and improves computational efficiency. This streamlined feature set allows the classifier to focus on the most relevant patterns, enhancing its predictive accuracy and performance metrics.

Machine learning (ML) algorithms

Machine Learning (ML) algorithms are computational techniques used to build models that can learn patterns from data and make predictions or decisions without being explicitly programmed. These algorithms are crucial for tasks such as detection attacks in IDS for Wireless Sensor Networks (WSNs).

Decision Tree Classifier (DTC)

A Decision Tree Classifier (DTC) is a tree-like structure where each internal node represents a feature (or attribute), each branch represents a decision rule, and each leaf node represents a class label. The algorithm splits the dataset into subsets based on feature values, recursively creating branches to partition the data into homogenous subsets.

The decision tree algorithm uses the following recursive rule for splitting:

$$\text{Best Split} = \arg\max_{f \in F} \text{Information Gain}(f) \tag{10}$$

where

$$\text{Information Gain}(f) = H(S) - \sum_{v \in V(f)} \frac{|S_v|}{|S|} H(S_v) \tag{11}$$

Here, $H(S)$ is the entropy of the dataset, and S_v is the subset of data for each value v of the feature f.

DTC is easy to interpret and visualize, making it suitable for IDS models. It effectively handles both numerical and categorical data and is less prone to overfitting compared to other algorithms, especially when pruned.

Random Forest Classifier (RFC)

A Random Forest Classifier (RFC) is an ensemble learning method that builds multiple decision trees using different subsets of the data and features. Each tree makes an independent prediction, and the final output is determined by aggregating the predictions (usually by majority voting for classification).

The output of RFC can be represented as:

$$\hat{y}_{\text{RFC}} = \frac{1}{T} \sum_{t=1}^{T} \hat{y}_t \tag{12}$$

where T is the total number of trees, and $\hat{y}_t$ is the prediction from the t-th tree.

RFC is robust to overfitting, handles large datasets well, and improves prediction accuracy by reducing variance through bagging. It is also less sensitive to noisy data and outliers.

CatBoost Classifier (CBC)

The CatBoost Classifier (CBC) is an advanced gradient boosting algorithm designed for classification tasks. It is based on the gradient boosting framework but has several key optimizations to improve performance, reduce overfitting, and handle categorical features efficiently. CatBoost stands for "Categorical Boosting" and is particularly useful when the dataset contains categorical variables, as it can process them directly without the need for manual preprocessing, such as one-hot encoding.

The general form of the gradient boosting equation for CatBoost is as follows:

$$F(x) = \sum_{m=1}^{M} \eta_m h_m(x) \tag{13}$$

where $F(x)$ is the prediction function. η_m is the learning rate for the m-th boosting round. $h_m(x)$ is the weak learner (usually a decision tree) fitted in the m-th iteration. M is the total number of boosting iterations.

In CatBoost, the decision trees $h_m(x)$ are built using a gradient-based optimization process that minimizes a loss function, typically the log-loss or binary cross-entropy in classification problems.

With techniques like ordered boosting and regularization, CatBoost reduces overfitting and ensures better generalization on unseen data. Moreover, it requires fewer hyperparameters, making it faster to train compared to other gradient boosting algorithms, and is robust against noisy data and outliers.

Extreme Gradient Boosting Classifier (XGBC)
Extreme Gradient Boosting (XGBC) is an advanced implementation of gradient boosting that optimizes model training by using decision trees in a boosting framework. XGBC builds trees sequentially, where each tree corrects the errors of the previous one. It uses regularization to prevent overfitting and improve generalization.

The final prediction in XGBC is given by:

$$\hat{y} = \sum_{t=1}^{T} \alpha_t f_t(x) \tag{14}$$

where α_t is the weight of the t-th tree and $f_t(x)$ is the prediction of the t-th tree.

XGBC is highly efficient and scalable, providing high predictive accuracy. It is robust against overfitting due to its regularization mechanism and is widely used in IDS due to its ability to handle large datasets and complex relationships.

LightGBM Classifier (LGBC)
LightGBM (Light Gradient Boosting Machine) is a gradient boosting framework that uses a histogram-based approach for faster training. It builds decision trees leaf-wise (as opposed to level-wise), which helps reduce the number of splits and improves computational efficiency.

The prediction in LGBC can be written as:

$$\hat{y} = \sum_{t=1}^{T} \alpha_t f_t(x) \tag{15}$$

Similar to XGBC, but with improved efficiency and speed due to the histogram-based approach.

LGBC is faster and more memory-efficient than traditional gradient boosting algorithms, making it suitable for large-scale IDS datasets. It can handle categorical features directly and achieves high accuracy in a shorter training time.

Histogram-based Gradient Boosting Classifier (HGBC)
Histogram-based Gradient Boosting (HGBC) builds trees using gradient boosting and bins continuous features into histograms, which reduces the number of calculations during tree construction. This makes it efficient, especially for large datasets.

Similar to LightGBM, HGBC's prediction is given by:

$$\hat{y} = \sum_{t=1}^{T} \alpha_t f_t(x) \tag{16}$$

where $f_t(x)$ is the decision function of the t-th tree, and α_t is the corresponding weight.

HGBC offers high accuracy and efficiency, with a focus on memory usage optimization, making it well-suited for IDS in resource-constrained environments. It allows for fast model training and is capable of handling high-dimensional data with ease.

Result analysis

Result analysis plays a crucial role in evaluating the effectiveness of IDS built using ML models. It involves assessing the ability of the model to accurately classify intrusion events while ensuring minimal false positives and false negatives. This analysis typically includes metrics such as accuracy, precision, recall, f1-score, and area under the curve (AUC), which offer insights into the model's overall performance and robustness. The results obtained help in comparing different models and selecting the one with the best performance for deployment in real-world scenarios.

Experiment setup

The experiments for the IDS model were conducted on a system equipped with Ubuntu 20.04 LTS as the operating system, ensuring compatibility with various ML libraries. The system was configured with 16 GB of RAM and a 512 GB ROM, allowing efficient handling of large datasets. For processing, an Intel Core i7-9700K processor was used, coupled with a 1 TB SSD for faster data access and a 2 TB HDD for additional storage. To support the execution of ML models, several Python libraries were utilized, including `scikit-learn` for traditional algorithms, `xgboost` for extreme gradient boosting, `lightgbm` for Light Gradient Boosting and `pandas`, `numpy` along with `matplotlib` for implementing data reading, numerical computing, and visualization on Jupyter Notebook, Anaconda Navigator environment. This setup provided a robust environment to conduct model training, testing, and performance evaluation effectively.

Performance metrics

- **Confusion Matrix**: The confusion matrix (Table 5) displays the performance of a classification model by showing the counts of True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). The matrix can be used to compute various evaluation metrics.
- Accuracy:

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN} \tag{17}$$

- Precision:

$$Precision = \frac{TP}{TP + FP} \tag{18}$$

- Sensitivity (Recall):

$$Sensitivity = \frac{TP}{TP + FN} \tag{19}$$

- Specificity:

$$\text{Specificity} = \frac{TN}{TN + FP} \tag{20}$$

- F1-Score:

$$F1\text{-}score = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{21}$$

- Cohen's Kappa (Kappa):

$$\kappa = \frac{P_o - P_e}{1 - P_e} \tag{22}$$

where
- P_o = observed agreement (the proportion of correct predictions).
- P_e = expected agreement (the proportion of agreement that would be expected by chance).

- Matthews Correlation Coefficient (MCC):

	Actual positive	Actual negative
Predicted positive	TP	FP
Predicted negative	FN	TN

Table 5. Confusion matrix.

$$MCC = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}} \tag{23}$$

- ROC-AUC:

$$\text{AUC} = \int_0^1 TPR(FPR)\, dFPR \tag{24}$$

-
  - $TruePositiveRate(TPR) = \frac{TP}{TP+FN}$
  - $FalsePositiveRate(FPR) = \frac{FP}{FP+TN}$

- Mean Absolute Error—MAE:

$$MAE = \sum_{i=1}^{n} \frac{predicted(i) - actual(i)}{n} \tag{25}$$

- Mean Squared Error—MSE :

$$MSE = \sum_{i=1}^{n} \frac{(predicted(i) - actual(i))^2}{n} \tag{26}$$

- Root Mean Squared Error—RMSE:

$$RMSE = \sqrt{\sum_{i=1}^{n} \frac{(predicted(i) - actual(i))^2}{n}} \tag{27}$$

Here, n represents the total number of values.

Performance analysis

In our experiments, various data balancing techniques, including SMOTE TomekLink (STL), Generative Adversarial Networks (GAN), and K-Means Sampling (KMS) and Principal Component Analysis (PCA) for efficiently reducing the high-dimensional feature space while retaining the most significant information. Metrics such as accuracy, precision, sensitivity, specificity, F1 score, kappa, MCC, ROC-AUC, MAE, MSE, and RMSE were used to evaluate six models: Decision Tree Classifier (DTC), Random Forest Classifier (RFC), CatBoost Classifier (CBC), XGBoost Classifier (XGBC), LightGBM Classifier (LGBC), and HistGradientBoosting Classifier (HGBC). Among these combinations, the KMS+PCA+RFC model demonstrated the best performance, achieving superior accuracy in detecting attacks effectively. This highlights the importance of selecting the appropriate data balancing technique while retaining the most significant information and model for improving the accuracy and reliability of IDS systems.

Table 6 shows the performance of various ML models on the WSN dataset across three experimental setups: STL, GAN, and KMS. Among these, the KMS+PCA+RFC combination consistently outperforms others, achieving the highest metrics across all categories, including accuracy (99.94%), precision (99.94%), sensitivity (99.94%), specificity (99.99%), f1-score (99.94%), Kappa (99.93%), MCC (99.93%), and ROC-AUC (99.99%). Furthermore, KMS+PCA+RFC exhibits the lowest error rates, with minimal MAE (0.16), MSE (0.53), and RMSE (7.30). These results highlight the effectiveness and robustness of the proposed KMS+PCA+RFC model compared to other techniques such as DTC, CBC, XGBC, LGBC, and HGBC across different data balancing methods.

Table 7 illustrates the performance analysis of various hybrid machine learning models applied to the TON-IoT dataset under three data balancing techniques: STL, GAN and KMN. RFC consistently achieved the highest performance across all setups, with near-perfect accuracy, specificity, and ROC-AUC scores, followed by XGBC, which also exhibited strong results, especially under the KMS technique. Models like CBC and HGBC showed reliable performance, with slight improvements using GAN balancing, while LGBC had the lowest metrics, including higher error rates. The analysis highlights RFC and XGBC as the most effective models for the TON-IoT dataset, demonstrating their robustness and scalability across different data balancing techniques. The KMS combined with the RFC achieved exceptional performance on the TON-IoT dataset, showcasing an accuracy of 99.97%, precision of 99.97%, sensitivity of 99.97%, specificity of 100.0%, F1 score of 99.97%, kappa of 99.97%,

DB+DR Tech.	Model	Accuracy	Precision	Sensitivity	Specificity	F1score	Kappa	MCC	ROC-AUC	MAE	MSE	RMSE
STL+PCA	DTC	99.31	99.31	99.31	99.83	99.31	99.14	99.14	99.57	1.86	5.47	23.40
	RFC	**99.65**	**99.65**	**99.65**	**99.91**	**99.65**	**99.56**	**99.56**	**99.99**	**0.98**	**2.95**	**17.19**
	CBC	97.98	98.03	97.98	99.50	97.98	97.48	97.49	99.91	4.58	12.33	35.12
	XGBC	98.32	98.37	98.32	99.58	98.32	97.91	97.92	99.94	4.03	11.13	33.37
	LGBC	97.68	97.74	97.68	99.42	97.67	97.10	97.12	99.85	5.20	13.86	37.23
	HGBC	97.18	97.26	97.18	99.29	97.17	96.47	96.50	99.84	5.78	14.65	38.27
GAN+PCA	DTC	99.80	99.80	99.80	99.95	99.80	99.74	99.74	99.87	0.47	1.35	11.63
	RFC	**99.86**	**99.86**	**99.86**	**99.97**	**99.86**	**99.83**	**99.83**	**99.98**	**0.31**	**0.87**	**9.35**
	CBC	99.75	99.75	99.75	99.94	99.75	99.69	99.69	100.0	0.43	1.06	10.30
	XGBC	99.77	99.77	99.77	99.94	99.77	99.72	99.72	100.0	0.42	1.06	10.27
	LGBC	99.72	99.72	99.72	99.93	99.72	99.65	99.65	99.77	0.65	1.82	13.48
	HGBC	99.67	99.67	99.67	99.92	99.67	99.58	99.58	99.99	0.55	1.25	11.20
KMS+PCA	DTC	99.89	99.89	99.89	99.97	99.89	99.87	99.87	99.93	0.29	0.90	9.50
	RFC	**99.94**	**99.94**	**99.94**	**99.99**	**99.94**	**99.93**	**99.93**	**99.99**	**0.16**	**0.53**	**7.30**
	CBC	99.89	99.89	99.89	99.97	99.89	99.86	99.86	100.0	0.23	0.66	8.12
	XGBC	99.93	99.93	99.93	99.98	99.93	99.92	99.92	100.0	0.17	0.52	7.21
	LGBC	99.74	99.74	99.74	99.94	99.74	99.68	99.68	99.98	0.47	1.18	10.86
	HGBC	99.91	99.91	99.91	99.98	99.91	99.89	99.89	99.99	0.21	0.64	8.02

Table 6. Performance analysis of ML models on WSN dataset. Significant values are in bold.

DB+DR Tech.	Model	Accuracy	Precision	Sensitivity	Specificity	F1score	Kappa	MCC	ROC-AUC	MAE	MSE	RMSE
STL+PCA	DTC	99.95	99.95	99.95	99.99	99.95	99.94	99.94	99.97	0.19	1.04	10.19
	RFC	**99.97**	**99.97**	**99.97**	**100.0**	**99.97**	**99.97**	**99.97**	**100.0**	**0.10**	**0.49**	**7.03**
	CBC	99.52	99.52	99.52	99.95	99.52	99.46	99.46	100.0	1.36	6.20	24.89
	XGBC	99.78	99.78	99.78	99.98	99.78	99.75	99.75	100.0	0.69	3.17	17.80
	LGBC	98.63	98.63	98.63	99.85	98.62	98.47	98.47	98.82	6.50	40.08	63.31
	HGBC	99.42	99.42	99.42	99.94	99.42	99.35	99.35	99.99	1.92	9.25	30.41
GAN+PCA	DTC	99.95	99.95	99.95	99.99	99.95	99.95	99.95	99.97	0.19	1.04	10.18
	RFC	**99.97**	**99.97**	**99.97**	**100.0**	**99.97**	**99.97**	**99.97**	**100.0**	**0.12**	**0.63**	**7.92**
	CBC	99.83	99.83	99.83	99.98	99.83	99.81	99.81	100.0	0.61	3.12	17.68
	XGBC	99.89	99.89	99.89	99.99	99.89	99.87	99.87	100.0	0.41	2.00	14.14
	LGBC	96.89	96.92	96.89	99.65	96.88	96.54	96.55	97.83	8.38	39.49	62.84
	HGBC	99.69	99.69	99.69	99.97	99.69	99.65	99.65	99.90	1.31	6.92	26.31
KMS+PCA	DTC	99.96	99.96	99.96	100.0	99.96	99.95	99.95	99.98	0.18	0.95	9.74
	RFC	**99.97**	**99.97**	**99.97**	**100.0**	**99.97**	**99.97**	**99.97**	**100.0**	**0.11**	**0.59**	**7.68**
	CBC	99.74	99.74	99.74	99.97	99.74	99.71	99.71	100.0	0.94	4.58	21.40
	XGBC	99.90	99.90	99.90	99.99	99.90	99.89	99.89	100.0	0.38	1.85	13.62
	LGBC	99.49	99.49	99.49	99.94	99.49	99.43	99.43	99.85	1.82	9.42	30.69
	HGBC	99.64	99.64	99.64	99.96	99.64	99.60	99.60	99.98	1.38	6.68	25.85

Table 7. Performance analysis of ML models on TON-IoT network dataset. Significant values are in bold.

and MCC of 99.97%. The model also attained a perfect ROC-AUC score of 100.0%, reflecting its ability to distinguish between classes. Furthermore, the error metrics were minimal, with an MAE of 0.11, MSE of 0.59, and RMSE of 7.68, highlighting its precision and robustness in predicting outcomes with high reliability.

The confusion matrix (Table 8) for the proposed KMS+PCA+RFC model on the WSN dataset demonstrates its outstanding performance in accurately classifying network attacks. For the Normal class, 68,114 instances were correctly classified, with minimal misclassifications: 21 as Grayhole, 2 as Blackhole, 2 as TDMA, and 56 as Flooding. Similarly, the Grayhole class had 67,575 correctly classified instances, with negligible misclassifications: 10 as Normal and 9 as Blackhole. The Blackhole class achieved near-perfect accuracy, with 68,249 instances correctly classified and only 5 misclassified as Grayhole. For the TDMA class, 68,188 instances were correctly identified, with minor errors: 86 as Normal and 2 as Grayhole. Finally, the Flooding class was accurately classified for 67,743 instances, with only 4 instances misclassified as Normal. These results underscore the high precision and robustness of the KMS+PCA+RFC model in distinguishing between different types of network behaviors and attacks.

	Normal	Grayhole	Blackhole	TDMA	Flooding
Normal	68114	21	2	2	56
Grayhole	10	67575	9	0	0
Blackhole	0	5	68249	0	0
TDMA	86	2	0	68188	0
Flooding	4	0	0	0	67743

Table 8. Confusion matrix of the proposed (KMS+PCA+RFC) model on WSN-DS.

	Normal	Scanning	Dos	Injection	Ddos	Password	Xss	Ransomware	Backdoor	Mitm
Normal	60074	1	2	1	5	26	21	5	0	0
Scanning	2	60043	2	0	1	0	0	0	0	0
Dos	7	0	59973	4	1	0	0	0	1	0
Injection	4	0	0	59981	10	1	0	0	0	0
Ddos	6	0	0	11	59983	0	1	0	0	0
Password	4	0	0	0	1	60085	0	0	0	0
Xss	20	0	0	1	1	1	60054	0	0	0
Ransomware	1	0	0	0	0	0	0	59835	0	0
Backdoor	1	0	0	0	0	0	0	0	59801	1
Mitm	7	0	0	0	0	0	3	0	0	60024

Table 9. Confusion matrix of the proposed (KMS+PCA+RFC) model on TON-IoT-network.

The confusion matrix (Table 9) for the proposed KMS+PCA+RFC model on the TON-IoT Network dataset demonstrates its remarkable accuracy across 10 classes of network activities and attacks. The Normal class achieved excellent classification with 60,074 correct predictions, and minimal misclassifications: 1 as Scanning, 2 as DoS, 1 as Injection, 5 as DDoS, 26 as Password, 21 as XSS, and 5 as Ransomware. The Scanning class also showed high accuracy, with 60,043 correctly classified instances and minor errors, such as 2 misclassified as Normal and 1 as DDoS. The DoS class was accurately classified for 59,973 instances, with minimal confusion: 7 as Normal, 4 as Injection, and 1 as Backdoor. Similarly, the Injection class had 59,981 accurate predictions, with slight misclassifications: 4 as Normal, 10 as DDoS, and 1 as Password. For the DDoS class, 59,983 instances were correctly classified, with minor errors: 6 as Normal, 11 as Injection, and 1 as XSS. The Password class was nearly perfect with 60,085 correct classifications, and only 1 misclassified as DDoS. The XSS class also performed well with 60,054 correct classifications, with minor misclassifications: 20 as Normal, 1 each as Injection and DDoS, and 1 as Password. The RansomwareThe confusion matrix for the proposed KMS+PCA+RFC model on the TON-IoT Network dataset demonstrates its remarkable accuracy across 10 classes of network activities and attacks. The Normal class achieved excellent classification with 60,074 correct predictions, and minimal misclassifications: 1 as Scanning, 2 as DoS, 1 as Injection, 5 as DDoS, 26 as Password, 21 as XSS, and 5 as Ransomware. The Scanning class also showed high accuracy, with 60,043 correctly classified instances and minor errors, such as 2 misclassified as Normal and 1 as DDoS. The DoS class was accurately classified for 59,973 instances, with minimal confusion: 7 as Normal, 4 as Injection, and 1 as Backdoor. Similarly, the Injection class had 59,981 accurate predictions, with slight misclassifications: 4 as Normal, 10 as DDoS, and 1 as Password. For the DDoS class, 59,983 instances were correctly classified, with minor errors: 6 as Normal, 11 as Injection, and 1 as XSS. The Password class was nearly perfect with 60,085 correct classifications, and only 1 misclassified as DDoS. The XSS class also performed well with 60,054 correct classifications, with minor misclassifications: 20 as Normal, 1 each as Injection and DDoS, and 1 as Password. The Ransomware class achieved 59,835 correct classifications, with negligible confusion: 1 misclassified as Normal. The Backdoor class saw 59,801 correctly classified instances, with 1 misclassified as MitM and 1 as Normal. Finally, the MitM class achieved strong performance with 60,024 accurate classifications and minimal errors: 7 as Normal and 3 as XSS. Overall, this confusion matrix highlights the exceptional capability of the KMS+PCA+RFC model to effectively differentiate between diverse network behaviors and attack types, with extremely low misclassification rates. class achieved 59,835 correct classifications, with negligible confusion: 1 misclassified as Normal. The Backdoor class saw 59,801 correctly classified instances, with 1 misclassified as MitM and 1 as Normal. Finally, the MitM class achieved strong performance with 60,024 accurate classifications and minimal errors: 7 as Normal and 3 as XSS.

Overall, both confusion matrices highlight the exceptional capability of the KMS+PCA+RFC model to effectively differentiate between diverse network behaviors and attack types, with extremely low misclassification rates.

The classification report (Table 10) for the proposed RFC (KMS+PCA+RFC) model on the WSN-DS dataset demonstrates exceptional performance across all five classes (Normal, Grayhole, Blackhole, TDMA, and Flooding), surpassing most other models evaluated in the study . The following key observations can be made: Normal Class: The RFC model achieved a precision of 99.85%, a recall of 99.88%, and an f1-score of 99.87%. These metrics indicate highly reliable identification of Normal network behavior with minimal false positives

Model	Class	Precision	Recall	F1-score	Model	Class	Precision	Recall	F1-score
DTC	Normal	99.77	99.75	99.76	XGBC	Normal	99.85	99.89	99.87
	Grayhole	99.92	99.93	99.93		Grayhole	99.95	99.94	99.95
	Blackhole	99.98	99.97	99.97		Blackhole	99.95	99.99	99.97
	TDMA	99.86	99.88	99.87		TDMA	99.99	99.87	99.93
	Flooding	99.94	99.94	99.94		Flooding	99.93	99.99	99.96
RFC	Normal	99.85	99.88	99.87	LGBC	Normal	99.66	99.68	99.67
	Grayhole	99.96	99.97	99.97		Grayhole	99.83	99.18	99.51
	Blackhole	99.98	99.99	99.99		Blackhole	99.80	99.94	99.87
	TDMA	100.00	99.87	99.93		TDMA	99.83	99.84	99.84
	Flooding	99.92	99.99	99.96		Flooding	99.48	99.95	99.71
CBC	Normal	99.85	99.84	99.84	HGBC	Normal	99.83	99.79	99.81
	Grayhole	99.91	99.78	99.85		Grayhole	99.90	99.93	99.92
	Blackhole	99.80	99.96	99.88		Blackhole	99.95	99.99	99.97
	TDMA	100.00	99.87	99.93		TDMA	99.94	99.88	99.91
	Flooding	99.90	99.99	99.94		Flooding	99.94	99.98	99.96

Table 10. Classification reports of (KMS+PCA+ML) models on WSN-DS.

or negatives. Grayhole Class: The Grayhole class exhibited outstanding classification metrics with a precision of 99.96%, a recall of 99.97%, and an f1-score of 99.97%. This demonstrates the model's robust ability to detect and classify Grayhole attacks accurately. Blackhole Class: The RFC model delivered near-perfect results for Blackhole attacks, achieving a precision of 99.98%, a recall of 99.99%, and an f1-score of 99.99%. These scores signify the model's capability to handle this attack type with precision and consistency. TDMA Class: The model achieved a perfect precision of 100.00%, a recall of 99.87%, and an f1-score of 99.93% for TDMA. This highlights the model's strength in minimizing false positives while maintaining high recall. Flooding Class: The Flooding class classification also showed excellent results, with a precision of 99.92%, a recall of 99.99%, and an f1-score of 99.96%. This indicates the model's ability to accurately identify Flooding attacks with almost no misclassification.

The RFC model's performance surpasses most of the other models (DTC, CBC, LGBC, HGBC, and XGBC) in terms of precision, recall, and f1-score across all classes. Notably, its metrics for Blackhole, Grayhole, and TDMA classes outperform all competing models. While some models like XGBC and HGBC also achieve competitive scores, the RFC model demonstrates consistently high performance with minimal variability across all classes, making it the most reliable option among the models tested. The RFC (KMS+PCA+RFC) model demonstrates superior classification capabilities across all network behavior and attack types on the WSN-DS dataset, making it an excellent choice for robust and precise intrusion detection in wireless sensor networks.

Table 11 compares the classification performance of various models on the TON-IoT Network dataset, evaluated across multiple attack categories. Key metrics such as Precision, Recall, and f1-score are provided for each class (Normal, Scanning, Dos, Injection, Ddos, Password, Xss, Ransomware, Backdoor, and Mitm). DTC: Achieves excellent performance across all classes, with nearly perfect metrics for Scanning, Ransomware, Backdoor, and Mitm (f1-score $\geq 99.99\%$). Slightly lower but still strong results for Normal (f1-score: 99.85%) and Xss (f1-score: 99.92%). XGBC: Strong overall performance but marginally lower than DTC in several classes (e.g., Normal with f1-score: 99.67%). Excels in Scanning, Ransomware, Backdoor, and Mitm, maintaining near-perfect metrics. RFC (Proposed Model): Outperforms other models with consistently high metrics across all classes. Achieves perfect scores for Scanning, Backdoor, and Mitm (Precision, Recall, and f1-score = 100%). Demonstrates robust generalization with a slightly lower f1-score for Normal (99.91%). LGBC: Metrics are notably lower compared to other models, especially for Normal (f1-score: 96.92%) and Backdoor (f1-score: 97.27%). Struggles with classes such as Password and Ddos, with performance below that of DTC and RFC. CBC: Strong performance with near-perfect scores for Backdoor and Mitm (f1-score = 99.99% and 99.94%, respectively). Slightly underperforms in Normal and Injection classes compared to DTC and RFC. HGBC: Competitive with CBC, achieving near-perfect scores for most attack classes (e.g., Backdoor: 100%, Ransomware: 99.94%). Marginally lower f1-scores for Normal (98.74%) and Ddos (99.45%).

RFC (Proposed Model) demonstrates the most consistent and superior performance across all classes, particularly excelling in Scanning, Backdoor, and Mitm categories. Models like LGBC show room for improvement, especially in Normal and Ddos detection. All models achieve high classification metrics, highlighting the effectiveness of the KMS+ML approach on the TON-IoT dataset. Conclusion: The RFC model, when integrated with the KMS data balancing technique with PCA for the dimension reduction process, proves to be the most effective among the tested models for classifying normal and attack classes in the TON-IoT dataset. Its ability to achieve perfect precision, recall, and f1-scores for critical attack types underlines its robustness and suitability for network intrusion detection.

Justification of KFCV

To address the effectiveness of k-fold cross-validation (KFCV) in mitigating model overfitting and variance compared to alternative strategies like stratified sampling (SS), we conducted a comprehensive analysis using the KMS+ML models on the WSN dataset. Results presented in Table 12 indicate that both approaches deliver

Model	Class	Precision	Recall	F1-score	Model	Class	Precision	Recall	F1-score
DTC	Normal	99.89	99.81	99.85	XGBC	Normal	99.78	99.56	99.67
	Scanning	99.98	99.99	99.99		Scanning	99.98	99.99	99.98
	Dos	99.97	99.98	99.97		Dos	99.97	99.92	99.94
	Injection	99.96	99.94	99.95		Injection	99.87	99.88	99.87
	Ddos	99.94	99.96	99.95		Ddos	99.81	99.85	99.83
	Password	99.94	99.99	99.96		Password	99.72	99.99	99.86
	Xss	99.92	99.92	99.92		Xss	99.89	99.86	99.87
	Ransomware	99.98	99.99	99.99		Ransomware	99.97	99.98	99.98
	Backdoor	100.0	100.0	100.0		Backdoor	100.0	99.99	100.0
	Mitm	99.99	99.98	99.99		Mitm	99.99	99.97	99.98
RFC	Normal	99.91	99.90	99.91	LGBC	Normal	97.64	96.21	96.92
	Scanning	100.0	99.99	100.0		Scanning	98.60	98.77	98.68
	Dos	99.99	99.98	99.99		Dos	99.90	99.84	99.87
	Injection	99.97	99.97	99.97		Injection	99.14	99.37	99.26
	Ddos	99.97	99.97	99.97		Ddos	98.08	99.49	98.78
	Password	99.95	99.99	99.97		Password	97.42	99.49	98.44
	Xss	99.96	99.96	99.96		Xss	99.57	98.41	98.98
	Ransomware	99.99	100.0	99.99		Ransomware	98.97	99.36	99.16
	Backdoor	100.0	100.0	100.0		Backdoor	98.69	95.89	97.27
	Mitm	100.0	99.98	99.99		Mitm	98.29	99.41	98.85
CBC	Normal	99.63	98.81	99.22	HGBC	Normal	99.20	98.28	98.74
	Scanning	99.89	99.96	99.93		Scanning	99.89	99.96	99.92
	Dos	99.86	99.88	99.87		Dos	99.85	99.88	99.87
	Injection	99.80	99.62	99.71		Injection	99.71	99.54	99.63
	Ddos	99.65	99.66	99.66		Ddos	99.28	99.61	99.45
	Password	99.13	99.96	99.54		Password	99.12	99.93	99.53
	Xss	99.62	99.62	99.62		Xss	99.44	99.36	99.40
	Ransomware	99.92	99.94	99.93		Ransomware	99.97	99.90	99.94
	Backdoor	100.0	99.99	99.99		Backdoor	100.0	99.99	100.0
	Mitm	99.92	99.97	99.94		Mitm	99.92	99.95	99.94

Table 11. Classification reports of (KMS+PCA+ML) models on TON-IoT network.

SS VS KFCV	DB Tech.	Model	Accuracy	Precision	Sensitivity	Specificity	F1score	Kappa	MCC	ROC-AUC	MAE	MSE	RMSE
SS	**KMS**	DTC	99.89	99.89	99.89	99.97	99.89	99.86	99.86	99.93	0.29	0.89	9.42
		RFC	**99.94**	**99.94**	**99.94**	**99.98**	**99.94**	**99.92**	**99.92**	**99.99**	**0.17**	**0.53**	**7.31**
		CBC	99.89	99.89	99.89	99.97	99.89	99.86	99.86	100	0.23	0.64	7.99
		XGBC	99.93	99.93	99.93	99.98	99.93	99.91	99.91	100	0.18	0.53	7.3
		LGBC	99.72	99.72	99.72	99.93	99.72	99.65	99.65	99.8	0.46	1.04	10.19
KFCV	**KMS**	DTC	99.89	99.89	99.89	99.97	99.89	99.87	99.87	99.93	0.29	0.9	9.5
		RFC	**99.94**	**99.94**	**99.94**	**99.99**	**99.94**	**99.93**	**99.93**	**99.99**	**0.16**	**0.53**	**7.3**
		CBC	99.89	99.89	99.89	99.97	99.89	99.86	99.86	100	0.23	0.66	8.12
		XGBC	99.93	99.93	99.93	99.98	99.93	99.92	99.92	100	0.17	0.52	7.21
		LGBC	99.72	99.72	99.72	99.93	99.72	99.65	99.65	99.77	0.65	1.82	13.48

Table 12. SS VS KFCV using KMS+ML models on WSN-DS. Significant values are in bold.

robust performance metrics, with KFCV and SS achieving comparable accuracy, precision, and F1 scores across models. However, KFCV (k=5) demonstrates slightly better consistency in minimizing mean absolute error (MAE) and root mean square error (RMSE), particularly for the RFC model, which achieved the highest performance metrics (99.94% accuracy, precision, and F1 score). The stratification inherent in both methods ensures class balance, but KFCV's iterative training on multiple folds enhances generalizability and reduces variance, making it a more robust choice for scenarios involving diverse or imbalanced datasets.

Potential impact of PCA on performance

The high dimensionality and heterogeneity of the WSN-DS dataset indeed necessitate the use of dimensionality reduction techniques to optimize model performance and interpretability. Principal Component Analysis (PCA), applied alongside the Kaiser Criterion, plays a pivotal role in this context. PCA reduces the dataset's dimensionality by transforming it into a set of orthogonal components that maximize variance. This approach eliminates redundancy, minimizes noise, and mitigates the curse of dimensionality, which can hinder efficient training of IDS. In our study, PCA was applied to WSN-DS, retaining components with eigenvalues greater than 1, as per the Kaiser Criterion. This ensured that only the most significant features were used for training, reducing computational overhead while preserving over 95% of the data's variance. By streamlining the feature space, PCA enabled faster training and inference without sacrificing model accuracy (99.94%), which achieved a remarkable ROC-AUC of 99.99% in combination with the Hybrid (KMS+PCA+RFC) framework (Tables 6 and 15). Additionally, PCA improved the model's generalization capability by focusing on key variations, reducing overfitting risks. The interpretability of results was further enhanced as PCA allowed us to identify the most impactful features in intrusion detection scenarios (Figs. 2, 3). By demonstrating the benefits of PCA and the Kaiser Criterion, we highlight how dimensionality reduction not only boosts IDS performance but also ensures scalability and reliability, which are critical for resource-constrained wireless sensor network environments.

Discussion

Our proposed model, which integrates KMeansSMOTE (KMS) with Principal Component Analysis (PCA) and Random Forest Classifier (RFC), achieves outstanding performance for Intrusion Detection Systems (IDS) in Wireless Sensor Networks (WSNs), particularly on the WSN-DS dataset. A comparative analysis with existing works is presented in Table 13, highlighting the significant improvements offered by our approach. Talukder et al.[23] achieved a remarkable accuracy, precision, recall, and f1-score of 99.92% using STL with an RFC-based model. Similarly, Karthikeyan et al.[20] reported an accuracy of 99.34% with their FA-ML approach, though their f1-score was slightly lower at 96.67%. Abinayaa et al.[24] employed LOA techniques with their LOA-Cb-C model, achieving 99.66% accuracy and 99.00% for precision, recall, and f1-score. In contrast, hybrid and neural models such as CNN+RNN by Salmi et al.[25] and Dontu et al.[26] demonstrated lower performance, with an accuracy of 96.86% and an f1-score of 82.29%. Vaishnavi et al.[27] used PCA with XGB, reaching an accuracy of 99.70% but did not report precision, recall, or f1-scores. Our proposed model outperforms these existing works across all evaluation metrics, with an unparalleled accuracy, precision, recall, and f1-score of 99.94%. This improvement underscores the efficacy of combining KMS with PCA for dimensionality reduction and RFC for classification, making it a robust and reliable solution for IDS in WSNs.

For the TON-IoT Network dataset, our proposed model, integrating KMS with PCA and RFC, demonstrates exceptional performance. As shown in Table 14, our approach achieves the highest accuracy, precision, recall, and f1-score, setting a new benchmark for IDS in IoT networks. Cao et al.[28] employed STL and RFC, achieving an accuracy of 98.87% with precision, recall, and f1-scores of 99.36%, 99.18%, and 99.18%, respectively. Similarly, Uddin et al.[21] reported comparable accuracy and an f1-score of 99.37% using their usfAD model, although they did not report recall. Hajla et al.[29] utilized STL and a Voting-based model, achieving a lower accuracy of 96.32%

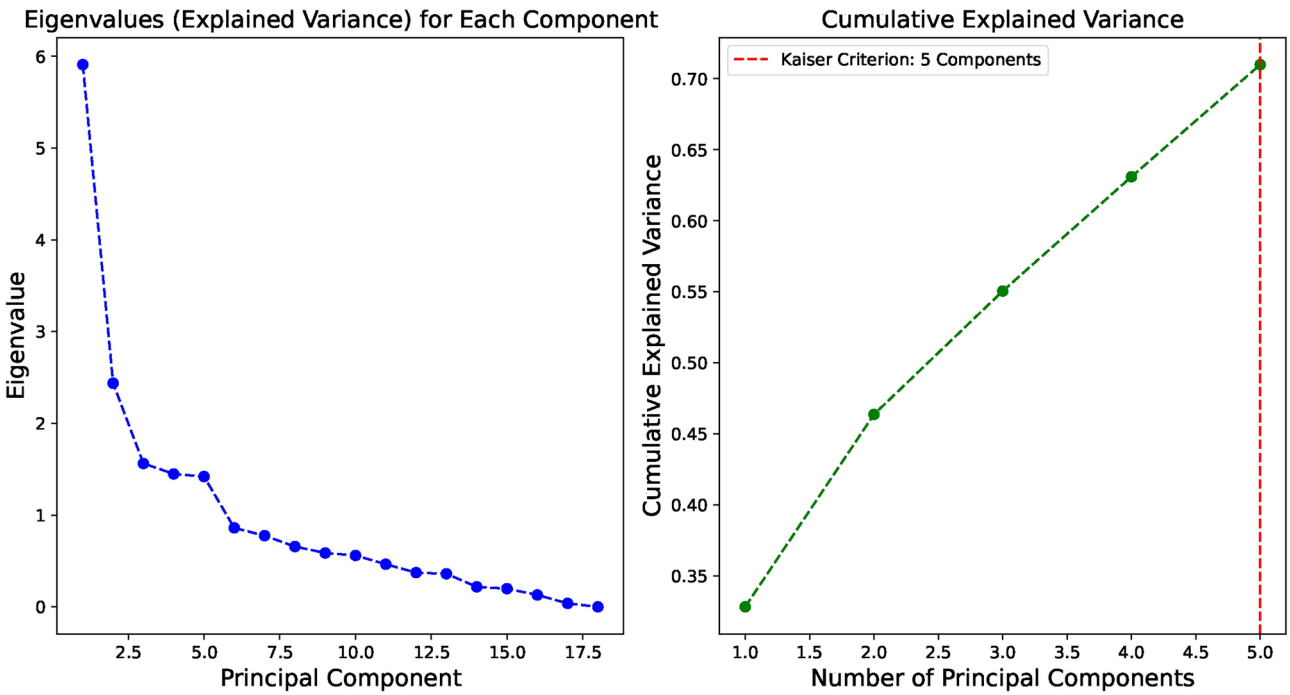

Fig. 2. Kaiser criterion for PCA on WSN-DS.

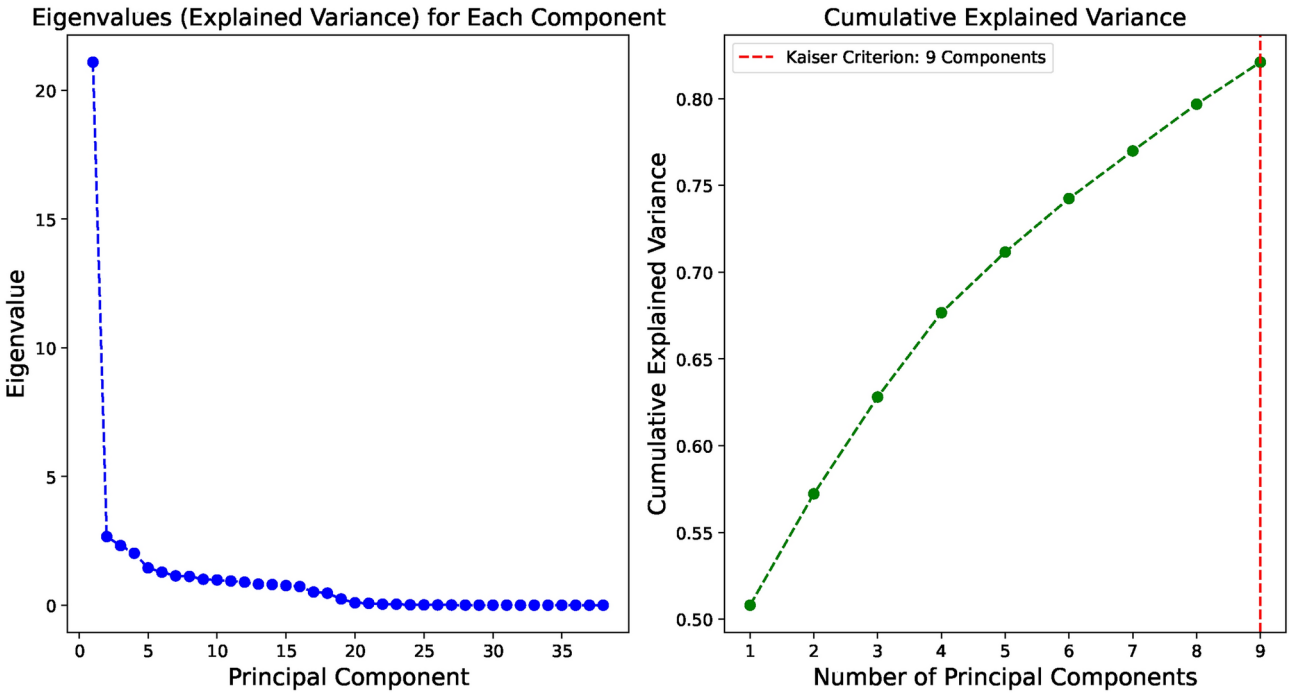

Fig. 3. Kaiser criterion for PCA on TON-IoT network.

Authors	DB Tech.	DD Tech.	Model	Accuracy	Precision	Recall	F1-score
Talukder et al.[23]	STL	–	RFC	99.92	99.92	99.92	99.92
Karthikeyan et al.[20]	–	–	FA-ML	99.34	-	98.36	96.67
Abinayaa et al.[24]	–	LOA	LOA-Cb-C	99.66	99.00	99.00	99.00
Salmi et al.[25]	–	–	CNN+RNN	96.86	85.19	85.29	82.29
Dontu et al.[26]	–	EABOA	Hybrid	96.86	85.19	85.29	82.29
Vaishnavi et al.[27]	–	PCA	XGB	99.70	–	–	–
Our Proposed	KMS	PCA	RFC	99.94	99.94	99.94	99.94

Table 13. Performance comparison of our proposed model with existing works on WSN-DS.

Authors	DB Tech.	DD Tech.	Model	Accuracy	Precision	Recall	F1-score
Cao et al.[28]	STL	–	RFC	98.87	99.36	99.18	99.18
Uddin et al.[21]	–	–	usfAD	98.87	99.36	–	99.37
Hajla et al.[29]	STL	–	Voting	96.32	93.11	84.55	88.63
Li et al.[30]	–	PCA	DT	77.62	67.34	48.67	45.53
Torre et al.[19]	–	–	DT	99.63	96.48	93.18	94.45
Uddin et al.[22]	–	–	OCC	98.94	–	–	98.94
Our Proposed	KMS	PCA	RFC	99.97	99.97	99.97	99.97

Table 14. Performance comparison of our proposed model with existing works on TON-IoT Network.

and an f1-score of 88.63%. Other models such as Li et al.[30], which applied PCA and Decision Tree (DT), reported significantly lower accuracy (77.62%) and an f1-score of 45.53%. Torre et al.[19] achieved an impressive accuracy of 99.63% with their DT model but lower precision (96.48%) and recall (93.18%). Uddin et al.[22] achieved a high accuracy of 98.94% using OCC but did not report precision or recall. Our proposed model surpasses all these works, achieving unparalleled metrics of 99.97% for accuracy, precision, recall, and f1-score. This demonstrates the robustness and efficiency of combining KMS with PCA for feature selection and RFC for classification, making it an ideal choice for IDS in the TON-IoT Network dataset.

The superior performance of the RFC in our study can be attributed to several factors related to the specific characteristics of the datasets used, namely WSN-DS and TON-IoT. Both of these datasets contain diverse attack

profiles and feature sets that align well with RFC's strengths in handling heterogeneous data. The RFC's ability to effectively manage both categorical and continuous features likely contributed to its strong performance on these datasets. In particular, on WSN-DS, RFC demonstrated high accuracy and robustness, yielding an accuracy of 99.94%, precision of 99.94%, and recall of 99.94%. Similarly, on the TON-IoT dataset, RFC also showed strong performance with an accuracy of 99.97%, precision of 99.97%, and recall of 99.97%. These results highlight the classifier's ability to adapt to different types of data and attack scenarios, reinforcing its effectiveness for intrusion detection in wireless sensor networks and IoT environments. While the RFC performed exceptionally well in these specific scenarios, its generalizability to other intrusion detection tasks with varying attack profiles will depend on how closely the features and data distributions in new datasets match those of WSN-DS and TON-IoT. Nonetheless, given RFC's robustness in these cases, we are confident that it could offer competitive performance in a variety of contexts.

Our methodology enhances the scalability and robustness of IDS in resource-constrained environments like WSNs by integrating an adaptive framework that leverages StandardScaler, KMS for data balancing, PCA with Kaiser Criterion for dimension reduction, and RFC. StandardScaler ensures feature standardization, which stabilizes the model and improves performance. KMeans SMOTE addresses class imbalance by generating synthetic samples of minority classes, thereby enhancing the model's ability to detect underrepresented attack patterns. PCA, applied using the Kaiser Criterion, reduces the dimensionality of the data while retaining only the most significant features, significantly lowering computational complexity. Finally, the RFC classifier is employed to effectively classify attacks with high accuracy and resilience, even in environments with limited resources. We validated the proposed approach on the WSN-DS and TON-IoT datasets, demonstrating its robust performance across diverse network conditions and attack profiles. This adaptive methodology ensures our model's scalability, efficiency, and resilience, making it well-suited for deployment in WSNs.

While advanced DL architectures, such as hybrid models, have demonstrated strong performance in domains like power systems (Graph wavelet CNN)[34] and in network attack detection (CNN+RNN)[25], we find that ML models are more compatible and effective in enhancing adaptability to complex attack patterns in WSNs than DL. DL models often require substantial computational resources, which may not be optimal in WSN environments with limited resources. Therefore, we focus on the use of the ML model, which offers a more efficient and effective solution for intrusion detection in such sensors and IOT netowrks.

Complexity analysis

The complexity analysis of our proposed model, which integrates KMeans-SMOTE (KMS) with PCA and RFC, is presented in Table 15. The analysis compares the training, prediction, and total time required for different machine learning models using various dataset preprocessing techniques. Experiment 1, which uses STL+PCA, shows that the RFC has the highest total time of 602.360 seconds, followed by the CBC at 381.434 seconds. In Experiment 2, using GAN+PCA, the total time is generally lower, with RFC taking 312.986 seconds, while the other models, such as DTC and XGBC, show significantly shorter training and prediction times. For Experiment 3, which applies KMS+PCA, our proposed model with RFC achieves a training time of 247.621 seconds and a prediction time of 3.468 seconds, resulting in a total time of 251.090 seconds. While RFC remains the model with the highest computational complexity, the integration of KMS+PCA significantly reduces the overall time compared to STL+PCA, making it more efficient. PCA-based preprocessing aids in reducing training times for most models, while maintaining high classification performance, which is crucial for real-time intrusion

Exp. No.	DB+DR Tech.	Model Name	Training Time (s)	Prediction Time (s)	Total Time (s)
1	STL+PCA	DTC	20.617	0.044	20.661
		RFC	597.124	5.236	602.360
		CBC	379.410	2.023	381.434
		XGBC	62.912	3.891	66.803
		LGBC	41.225	8.466	49.690
		HGBC	42.975	8.082	51.057
2	GAN+PCA	DTC	10.728	0.021	10.749
		RFC	310.220	2.766	312.986
		CBC	324.239	1.934	326.172
		XGBC	54.040	3.336	57.376
		LGBC	33.258	4.142	37.400
		HGBC	34.322	6.346	40.668
3	**KMS+PCA**	DTC	13.432	0.032	13.464
		RFC	**247.621**	**3.468**	**251.090**
		CBC	490.615	3.331	493.946
		XGBC	74.303	3.428	77.732
		LGBC	49.744	6.251	55.995
		HGBC	64.503	7.984	72.486

Table 15. Complexity analysis of ML models on WSN-DS. Significant values are in bold.

Exp. No.	DB. Tech.	Model Name	Training Time (s)	Prediction Time (s)	Total Time (s)
1	STL+PCA	DTC	36.745	0.052	36.796
		RFC	858.795	6.760	865.554
		CBC	1697.640	2.969	1700.610
		XGBC	91.203	5.297	96.500
		LGBC	70.621	28.021	98.642
		HGBC	61.308	8.388	69.696
2	GAN+PCA	**DTC**	36.051	0.042	36.093
		RFC	735.018	5.315	740.333
		CBC	1675.049	3.070	1678.118
		XGBC	85.460	4.856	90.317
		LGBC	63.061	23.389	86.450
		HGBC	57.499	8.210	65.709
3	**KMS+PCA**	DTC	40.307	0.062	40.369
		RFC	**887.191**	**7.347**	**894.537**
		CBC	1700.508	2.904	1703.411
		XGBC	90.031	5.042	95.072
		LGBC	74.114	28.524	102.638
		HGBC	47.937	6.400	54.336

Table 16. Complexity analysis of ML models on TON-IoT-Network. Significant values are in bold.

detection applications in IoT networks. These results highlight the trade-off between computation time and performance, with our proposed model offering an optimal balance for efficient and accurate predictions.

The complexity analysis of various ML models on the TON-IoT-Network dataset is presented in Table 16. This analysis evaluates the training, prediction, and total time required for each model under different preprocessing techniques. Experiment 1, using STL+PCA, reveals that the RFC has the longest total time of 865.554 seconds, followed by the CBC at 1700.610 seconds. Experiment 2, utilizing GAN+PCA, shows a reduction in total times for most models, with RFC taking 740.333 seconds. For Experiment 3, applying KMS+PCA, the RFC model again takes the longest time, totaling 894.537 seconds, but this is still efficient compared to the other models. Specifically, the DTC maintains a lower time requirement, with 40.369 seconds for total time. Although KMS+PCA preprocessing adds some time overhead, it helps in improving classification performance, as demonstrated by the shorter prediction times for most models compared to STL+PCA. This indicates that the application of KMS+PCA improves data quality for training, leading to better model efficiency and performance, while still maintaining reasonable computational costs. The complexity analysis highlights the trade-offs between model accuracy and time efficiency, with the proposed model (KMS+PCA+RFC) providing a balance of high performance and manageable computational demands for real-time applications in IoT-based intrusion detection systems.

The computational trade-offs and scalability of employing KMS for real-time intrusion detection in resource-constrained WSN environments were evaluated using the TON-IoT dataset. Results in Tables 6, 7 and 15, 16 demonstrate that KMS significantly outperforms STL and GAN in both computational efficiency and model performance respectively. The proposed model (KMS+PCA+RFC) achieved the lowest total time (251.090 seconds) compared to STL (602.360 seconds) and GAN (312.986 seconds) while delivering superior performance, including 99.94% accuracy, precision, recall, and F1 score in WSN. Similary on TON-IoT Network, achieved the lowest total time (8.94.537 seconds) compared to STL (865.554 seconds) and GAN (740.333 seconds) while delivering superior performance, including 99.97% accuracy, precision, recall, and F1 score. These results highlight the method's ability to correctly identify attacks with minimal computational overhead, making it ideal for large-scale and dynamic datasets. The balance between reduced computational cost and enhanced detection performance confirms its scalability and suitability for resource-constrained WSNs.

Practical implementation of IDS in real-world scenarios

Our research addresses the dynamic nature of WSN environments and evolving attack patterns by presenting a robust and adaptable intrusion detection framework. We employ advanced data processing techniques, including StandardScaler for feature standardization, KMeans SMOTE for handling class imbalance, and PCA with the Kaiser Criterion for dimensionality reduction, to efficiently process diverse attack scenarios. By validating our approach on benchmark datasets such as WSN-DS and TON-IoT, we demonstrate its effectiveness in real-world scenarios. Furthermore, the lightweight design of our machine learning models ensures compatibility with resource-constrained WSN environments, offering a scalable and efficient solution for real-time intrusion detection.

Addressing the adaptability to various attack types

The inherent limitations of KMS in certain scenarios, such as handling highly diverse attack types or adapting to varying network topologies. To address these challenges, we evaluated its performance alongside other

techniques, such as GAN-based oversampling and STL methods, to compare their effectiveness (Table 6). Our results demonstrate that KMS provides superior adaptability to diverse attack types by effectively balancing class distributions while maintaining computational efficiency (Table 15). Additionally, the integration of PCA with the Kaiser Criterion ensures dimensionality reduction tailored to the network's topology, further enhancing the robustness of our intrusion detection system (Fig. 2). This combination makes our framework versatile and effective in adapting to the dynamic nature of WSNs.

Addressing the continuous learning and adaptation to emerging threats in WSNs

Our research addresses the crucial aspect of continuous learning and adaptation by designing a flexible and robust intrusion detection framework tailored to the dynamic nature of WSNs. The integration of KMS ensures adaptability to evolving attack patterns by effectively addressing class imbalances, thereby enhancing the model's capacity to detect underrepresented threats. PCA with the Kaiser Criterion enables dynamic dimensionality reduction, focusing on the most critical features as data characteristics change, which optimizes computational efficiency. Additionally, the use of RFC provides high accuracy and resilience across diverse network conditions and attack profiles. By validating the methodology on benchmark datasets like WSN-DS and TON-IoT, we demonstrate its ability to adapt to emerging threats and varying scenarios, ensuring its scalability and practical applicability in real-world WSN environments.

Conclusion

In this research, we proposed a novel intrusion detection system (IDS) for wireless sensor networks (WSNs) by integrating K-means Smote (KMS) with Principal Component Analysis (PCA) and Random Forest Classifier (RFC). Our approach aimed at improving the performance and efficiency of IDS by addressing the challenges of class imbalance and dimensionality reduction in real-time applications. The experimental results demonstrated that our proposed model outperforms existing state-of-the-art models, achieving higher accuracy, precision, recall, and f1-score across different datasets, including WSN-DS and TON-IoT Network. The comparative analysis showed that KMS+PCA+RFC delivers superior classification performance, with the ability to effectively handle large-scale datasets and imbalanced classes. Moreover, the complexity analysis revealed that our model strikes a balance between classification accuracy and computational efficiency, making it suitable for deployment in resource-constrained environments such as WSNs. The time efficiency of our model was competitive, showing minimal increases in training and prediction times compared to other models. Overall, our research contributes to advancing IDS in WSNs, providing a robust and scalable solution for intrusion detection in IoT-based networks, and offering valuable insights for future work in this domain, particularly in optimizing the trade-off between model complexity and real-time application performance.

Despite the promising results of our proposed model, there are certain limitations that need to be addressed. First, the performance of the model relies on the quality of the feature extraction techniques, which may vary across different network conditions and types of attacks. Second, our approach does not yet account for the dynamic nature of wireless sensor networks, where environmental factors and changing network conditions could affect the model's privacy and accuracy.

For future work, we aim to explore adaptive feature selection methods that can dynamically adjust based on real-time network conditions. Furthermore, we plan to investigate federated learning techniques to improve the privacy of our model in large-scale, distributed WSN environments.

Data Availability

The selected datasets are sourced from free and open-access sources such as: WSN-DS dataset https://www.kaggle.com/datasets/bassamkasasbeh1/wsnds and TON-IoT Network dataset https://research.unsw.edu.au/projects/toniot-datasets.

Received: 11 August 2024; Accepted: 15 January 2025
Published online: 07 February 2025

References

1. Kumari, S. & Tyagi, A. K. Wireless sensor networks: An introduction. In *Digital Twin and Blockchain for Smart Cities*, 495–528 (2024).
2. Sharma, K. & Shivandu, S. K. Integrating artificial intelligence and internet of things (IoT) for enhanced crop monitoring and management in precision agriculture. *Sens. Int.* **5**, 100292 (2024).
3. Oztoprak, A., Hassanpour, R., Ozkan, A. & Oztoprak, K. Security challenges, mitigation strategies, and future trends in wireless sensor networks: A review. *ACM Comput. Surv.* (2024).
4. Bukhowah, R., Aljughaiman, A. & Rahman, M. H. Detection of dos attacks for IoT in information-centric networks using machine learning: Opportunities, challenges, and future research directions. *Electronics* **13**(6), 1031 (2024).
5. Pathak, V. et al. A secure and lightweight trust evaluation model for enhancing decision-making in resource-constrained industrial wsns. *Sci. Rep.* **14**(1), 28162 (2024).
6. Hussain, S., He, J., Zhu, N., Mughal, F. R., Hussain, M. I., Algarni, A. D., Ahmad, S., Zarie, M. M. & Ateya, A. A. An adaptive intrusion detection system for WSN using reinforcement learning and deep classification. *Arab. J. Sci. Eng.*, 1–15 (2024).
7. Abdulganiyu, O. H., Tchakoucht, T. A. & Saheed, Y. K. Towards an efficient model for network intrusion detection system (IDS): systematic literature review. *Wirel. Netw.* **30**(1), 453–482 (2024).
8. Sadia, H., Farhan, S., Haq, Y. U., Sana, R., Mahmood, T., Bahaj, S. A. O. & Rehman, A. Intrusion detection system for wireless sensor networks: A machine learning based approach. *IEEE Access* (2024).
9. Kikissagbe, B. R. & Adda, M. Machine learning-based intrusion detection methods in IoT systems: A comprehensive review. *Electronics* **13**(18), 3601 (2024).
10. Mohite, R. & Ouarbya, L. Interpretable anomaly detection: A hybrid approach using rule-based and machine learning techniques. In *2024 IEEE 9th International Conference for Convergence in Technology (I2CT)*, 1–10 (2024). IEEE.

11. Sarker, I. H., Janicke, H., Ferrag, M. A. & Abuadbba, A. Multi-aspect rule-based AI: Methods, taxonomy, challenges and directions toward automation, intelligence and transparent cybersecurity modeling for critical infrastructures. *Internet Things* **25**, 101110 (2024).
12. Li, Y., Wei, X., Li, Y., Dong, Z. & Shahidehpour, M. Detection of false data injection attacks in smart grid: A secure federated deep learning approach. *IEEE Trans. Smart Grid* **13**(6), 4862–4872 (2022).
13. Urmi, W. F. et al. A stacked ensemble approach to detect cyber attacks based on feature selection techniques. *Int. J. Cognit. Comput. Eng.* **5**, 316–331 (2024).
14. Talukder, M. A. et al. A dependable hybrid machine learning model for network intrusion detection. *J. Inf. Secur. Appl.* **72**, 103405 (2023).
15. Talukder, M. A. et al. Machine learning-based network intrusion detection for big and imbalanced data using oversampling, stacking feature embedding and feature extraction. *J. Big Data* **11**(1), 1–44 (2024).
16. Talukder, M. A., Hossen, R., Uddin, M. A., Uddin, M. N. & Acharjee, U. K. Securing transactions: A hybrid dependable ensemble machine learning model using IHT-LR and grid search. *Cybersecurity* **7**(1), 32 (2024).
17. Talukder, M. A., Khalid, M. & Uddin, M. A. An integrated multistage ensemble machine learning model for fraudulent transaction detection. *J. Big Data* **11**(1), 168 (2024).
18. Uddin, M. A., Aryal, S., Bouadjenek, M. R., Al-Hawawreh, M. & Talukder, M. A. Hierarchical classification for intrusion detection system: Effective design and empirical analysis. arXiv preprint arXiv:2403.13013 (2024).
19. Torre, D., Chennamaneni, A., Jo, J., Vyas, G. & Sabrsula, B. Towards enhancing privacy-preservation of a federated learning cnn intrusion detection system in IoT: Method and empirical study. *ACM Trans. Softw. Eng. Methodol.* (2024).
20. Karthikeyan, M., Manimegalai, D. & RajaGopal, K. Firefly algorithm based WSN-IoT security enhancement with machine learning for intrusion detection. *Sci. Rep.* **14**(1), 231 (2024).
21. Uddin, M. A., Aryal, S., Bouadjenek, M. R., Al-Hawawreh, M. & Talukder, M. A. usfAD based effective unknown attack detection focused IDS framework. *Sci. Rep.* **14**(1), 29103 (2024).
22. Uddin, M. A., Aryal, S., Bouadjenek, M. R., Al-Hawawreh, M. & Talukder, M. A. A dual-tier adaptive one-class classification IDS for emerging cyberthreats. *Comput. Commun.* **229**, 108006 (2025).
23. Talukder, M. A., Sharmin, S., Uddin, M. A., Islam, M. M. & Aryal, S. MLSTL-WSN: Machine learning-based intrusion detection using smotetomek in WSNs. *Int. J. Inf. Secur.* **23**(3), 2139–2158 (2024).
24. Abinayaa, S. S. et al. Securing the edge: Catboost classifier optimized by the lyrebird algorithm to detect denial of service attacks in internet of things-based wireless sensor networks. *Future Internet* **16**(10), 381 (2024).
25. Salmi, S. & Oughdir, L. Performance evaluation of deep learning techniques for dos attacks detection in wireless sensor network. *J. Big Data* **10**(1), 17 (2023).
26. Dontu, S., Vallabhaneni, R., Addula, S. R., Pareek, P. K. & Hussein, R. R. Enhanced adaptive butterfly optimizer based feature selection for protecting the data in industry based WSN. In *2024 International Conference on Intelligent Algorithms for Computational Intelligence Systems (IACIS)*, 1–6 (2024). IEEE.
27. Vaishnavi, S. & Reddy, R. Implementation of machine learning algorithms to detect dos attack in WSN. In *2024 5th International Conference on Circuits, Control, Communication and Computing (I4C)*, 386–391. (IEEE, 2024).
28. Cao, Z., Zhao, Z., Shang, W., Ai, S. & Shen, S. Using the TON-IoT dataset to develop a new intrusion detection system for industrial IoT devices. *Multimedia Tools Appl.* 1–29 (2024).
29. Hajla, S., Ennaji, E., Maleh, Y. & Mounir, S. Enhancing IoT network defense: advanced intrusion detection via ensemble learning techniques. *Indones. J. Electr. Eng. Comput. Sci.* **35**(3), 2010–2020 (2024).
30. Li, J., Othman, M. S., Chen, H. & Yusuf, L. M. Optimizing IoT intrusion detection system: Feature selection versus feature extraction in machine learning. *J. Big Data* **11**(1), 36 (2024).
31. Almomani, I., Al-Kasasbeh, B., Al-Akhras, M. et al. Wsn-ds: A dataset for intrusion detection systems in wireless sensor networks. *J. Sens.* **2016** (2016)
32. Moustafa, N. A new distributed architecture for evaluating AI-based security systems at the edge: Network ton_iot datasets. *Sustain. Cities Soc.* **72**, 102994 (2021).
33. Li, Y., Cao, J., Xu, Y., Zhu, L. & Dong, Z. Y. Deep learning based on transformer architecture for power system short-term voltage stability assessment with class imbalance. *Renew. Sustain. Energy Rev.* **189**, 113913 (2024).
34. Qu, Z. et al. Localization of dummy data injection attacks in power systems considering incomplete topological information: A spatio-temporal graph wavelet convolutional neural network approach. *Appl. Energy* **360**, 122736 (2024).

Acknowledgements
The authors extend their appreciation to Umm Al-Qura University, Saudi Arabia for funding this research work through grant number: 25UQU4290340GSSR01.

Author contributions
Md.A.T.: Conceptualization, Methodology, Software, Visualization, Investigation, Formal analysis, Supervision, Validation, Writing—Original draft preparation, Writing—Reviewing and Editing. M.K.: Visualization, Validation, Investigation and Writing-Review & editing. N.S.: Visualization, Validation, Formal Analysis, Investigation and Writing-Review & editing.

Funding
This research work was funded by Umm Al-Qura University, Saudi Arabia under grant number: 25UQU4290340GSSR01.

Declarations

Competing interests
The authors declare no competing interests.

Additional information
Correspondence and requests for materials should be addressed to M.A.T., M.K. or N.S.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

© The Author(s) 2025